

PROJECT RETAINAI

AI-Driven Early Warning Attrition System • Technical Assignment

Problem Statement: A company's operations and timely client deliveries are heavily dependent on its employees. Unplanned, immediate resignations lead to uncompleted tasks, delayed projects, and client dissatisfaction. As an AI Engineer, the objective is to design a predictive solution that identifies which employees are highly likely to resign in the current or upcoming month, allowing the HR and management teams to establish timely contingency plans.

A) Technical Methodology: How to Achieve This in AI

To proactively manage employee attrition, we formulate this challenge as a **Supervised Machine Learning** task, specifically a **Binary Classification** problem. The AI model's goal is to estimate the probability that a given employee will resign within a 30 to 60-day window.

1. Data Acquisition & Aggregation

The system integrates multiple internal organizational data sources:

- **Demographics:** Tenure, age, department, and distance from the office.
- **Engagement & Sentiment:** Job satisfaction scores (from quarterly surveys), relationship ratings with managers, and last feedback sentiment.
- **Performance & Activity:** Average overtime logged per week, number of projects concurrently assigned, and months elapsed since the last promotion.
- **Compensation:** Current monthly income, salary growth percentile, and bonus histories.

2. Preprocessing & Feature Engineering

Raw data is cleaned and transformed to extract high-predictive-value indicators:

- Imputing missing values in survey records.
- Encoding categorical variables (e.g., Department, Job Role).
- Deriving dynamic behavioral metrics, such as `Ratio_Overtime_vs_Standard_Hours` and `Promotion_Lag_Index`.

3. Model Selection & Training

Using historical HR records (where resignation outcomes are known), we train robust classification algorithms. Preferred choices include ensemble tree-based models like **Random Forest**, **XGBoost**, or **LightGBM** due to their capability to handle non-linear relationships and mixed data types effectively.

4. Evaluation Metrics

Because missing a true potential resignation (False Negative) has a higher business cost than investigating a false alarm (False Positive), we prioritize **Recall (Sensitivity)** alongside the **F1-Score** to maintain balanced model performance.

AI Inference Loop: On the 1st of every month, the batch prediction pipeline runs for all active employees. Individuals flagged with an attrition probability score exceeding 0.70 are compiled into a dashboard for retention mitigation steps.

B) The 3-Stage of Problem Identification

A systematic framework ensures the AI implementation aligns seamlessly with corporate deliverables:

Stage	Definition & Focus	Project RetainAI Application
1. Business Objective	Identifying the core commercial/ operational pain point and defining the success criteria.	Goal: Mitigate project delivery delays. Success Metric: Provide a minimum of 4-6 weeks of lead time to arrange alternate task resources.
2. Data Feasibility	Analyzing whether the organization captures the necessary historical data to train the model.	Reviewing access to HR information systems (HRIS), monthly timesheets, exit interviews, and performance appraisal logs over the last 3-5 years.
3. AI Task Formulation	Translating the verified business objective into a standard machine learning target.	Formulating the project as a binary classifier targeting Is_Attrition (Yes=1, No=0) based on rolling monthly employee profiles.

C) Proposed Project Nomenclature

A few formal options designed for executive presentations:

- **Project RetainAI:** Emphasizes the strategic goal of optimizing talent retention.
- **EWAS (Early Warning Attrition System):** Highlights the technical warning and risk mitigation framework.
- **TalentPulse:** Focuses on continuous workforce health and engagement monitoring.

D) Illustrative Model Input / Output Dataset

The following table displays a representative sample of features used by the model (refer to the attached Excel file for the full 25-record simulation):

ID	Tenure (Mths)	Satisfaction (1-5)	Overtime (Hrs/Wk)	Months Since Promo	Income (INR)	Projects	Is_Attrition
EMP001	14	2	12	14	65,000	4	1 (Yes)
EMP002	48	5	2	6	1,20,000	2	0 (No)
EMP003	6	4	0	6	45,000	1	0 (No)
EMP004	32	1	15	24	85,000	5	1 (Yes)
EMP005	24	4	5	12	95,000	3	0 (No)